RootGap: system architecture

One student model feeds both the student app and the teacher dashboard. Rule-based and explainable, no live AI.

Student on a phone
own or shared phone, class > name > PIN

Student with no device
answers on a printed worksheet with a confidence column

Teacher
phone or laptop, teacher PIN

STUDENT APP (PWA, works offline in the GitHub copy)

Access
class, name, PIN

Learn
21 lessons, worked examples, check question

Solve
hints, 3 levels, language retry

Test
Boss Battle, probes prerequisite

Revise
spaced review of due questions

TEACHER DASHBOARD

Class plan
gap groups, activities, peer pairs

Evidence + override
Likely / Confirmed, confirm or dismiss

Signals
fragile list, language flag, revision queue

Paper entry
type answers from worksheets

Questions
add own questions + mistake tags

Students
add / remove, set PIN, class filter

ADAPTIVE ENGINE (rule-based, runs in the browser)

Mastery update
correct or wrong, weighted by confidence and hints

Root-gap tracing
wrong option > mistake tag > earliest weak skill in the chain

Difficulty + pace
levels 1 to 3, skip ahead, step-by-step or fast track

Language-gap check
wrong in English, retry in Telugu or Hindi

Test controller
probe prerequisite, end early, boss HP

Revision scheduler
wrong 1d, unsure 2d, okay 4d, sure 7d to 30d

CONTENT (data, not code)

Skill graph
fractions > negative numbers > linear > quadratic

Question bank
25 questions, level 1-3, every wrong option tagged

Lessons
21 lessons + 4 bridge lessons, English / Telugu / Hindi UI

Teacher questions
added in the app

STUDENT RECORDS AND SYNC

Student record
mastery per skill, answer log with confidence and hints, lesson progress, review schedule, language log

Browser storage
works with no internet

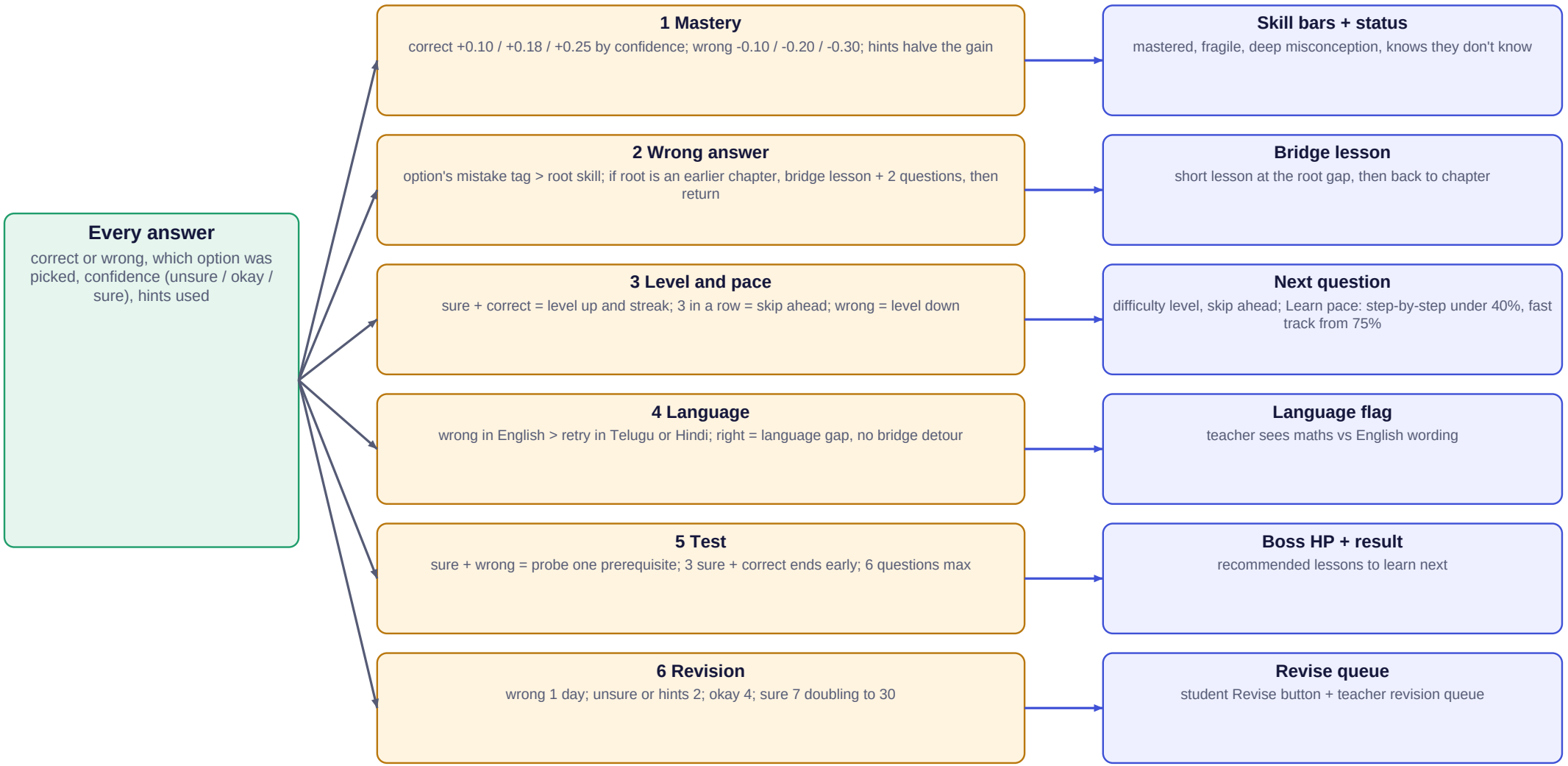
Shared database
syncs devices on the hosted link

Supabase (production)
planned for real classes and accounts

Red box = planned, not built. Demo PINs are a browser-side gate; production needs real accounts and server-side roles.

How RootGap decides what comes next

Every answer goes through the same rules. The teacher can see the evidence and overrule any diagnosis.



Then the teacher dashboard reads the same records

Groups students by shared root gap (not by marks), with a 10-minute no-device activity and peer pairs.

Evidence trail: question, option picked, mistake, confidence, hints. Labelled Likely (1-2 data points) or Confirmed (3+ or teacher-confirmed).

Teacher can confirm or dismiss a diagnosis, add questions, enter paper answers, and manage students. The system advises, the teacher decides.